FRC 2024 CRESCENDO — Top Robot Design Deep Dive

A study reference for effective CRESCENDO design & implementation, drawn from the Spectrum 3847 CAD Collection, team CAD/code releases (Onshape / GrabCAD), and each team's Chief Delphi reveal / CAD-release thread.

The CRESCENDO design landscape

The game

CRESCENDO is a NOTE game: robots collect NOTES (foam rings) off the floor or from the SOURCE and score them in the AMP or by shooting into the SPEAKER, then in the endgame CLIMB the CHAIN on the STAGE and optionally score a NOTE in the TRAP. Most elite designs centered on a fast, accurate speaker shooter (fixed or variable hood, sometimes a turret) fed by an under-the-bumper intake; many top teams treated the TRAP and HARMONY as optional, lower-value endgame choices and prioritized cycle speed.

How to use this document

These are the most thoroughly-documented CRESCENDO robots from the CAD-release set, each with real subsystem detail mined from the team's Chief Delphi reveal/release thread. For every team we capture archetype, the four subsystems (shooter / intake / indexer / drivetrain), vision, public CAD availability, and 'lessons for students'. ★ marks the curated study picks with the richest reproducible documentation. EPA / rank are pending a Statbotics API outage and will be backfilled when it recovers.

Profiled robots at a glance

	Team	Robot	Local EPA	Rec	Archetype	Public CAD/Design
★	**1678 Citrus Circuits**	(2024)	—	—	Elite fast NOTE cycle; first-ever CAD + code + STRATEGY releas	CD thread
★	**1690 Orbit**	Doppler	—	—	Remarkably compact, world-class fast-cycle speaker shooter	Onshape CAD
★	**118 Robonauts**	Twister	—	—	Highly complex shooter robot; CAD + code + tech binder + custo	CD thread
★	**8033 Highlander Robotics**	Banshee	—	—	Carefully-architected speaker shooter; SFR finalist, champs Mi	CD thread
★	**2910 Jack in the Bot**	Typhoon	—	—	Strong PNW shooter; CAD + code + tech binder, clone-friendly	CD thread
★	**4738 Patribots**	Terry the T. Rex	—	—	Requirements-driven shooter; standout technical binder	CD thread
★	**3061 Huskie Robotics**	(2024)	—	—	Open-Alliance, goals-first; full per-subsystem CAD + reusable	CD thread
★	**5940 BREAD**	Roti	—	—	Prototype-fast build; CAD + tech binder + ELECTRICAL binder +	CD thread
★	**6328 Mechanical Advantage**	(2024)	—	—	Open-Alliance software powerhouse; authors of AdvantageKit	CD thread
★	**2930 Sonic Squirrels**	Maestro	—	—	Strong PNW shooter robot (reveal-documented)	Onshape CAD
★	**125 NUTRONs**	MOMENTUM	—	—	Veteran NE team; bi-directional NOTE intake	Onshape CAD

★ = recommended study pick (downloadable CAD or a full technical binder + a clear design lesson). EPA shown is Local EPA (approx), computed from The Blue Alliance match data.

Team-by-team profiles

★ Study pick	**1678 Citrus Circuits** (2024) USA-CA	**EPA pending**

Elite fast NOTE cycle; first-ever CAD + code + STRATEGY release

SHOOTER (SPEAKER / AMP) Speaker-focused NOTE launcher (mechanism detail in the released CAD).	**INTAKE (NOTE)** NOTE intake feeding the launcher (see CAD).
INDEXER / FEED Feed path to the shooter (see CAD).	**DRIVETRAIN** Swerve.
VISION Unconfirmed (check CAD/code)	**PUBLIC CAD / DESIGN** CD CAD, code & strategy release

LESSONS FOR STUDENTS

- A top team releasing its STRATEGY and scouting whitepapers (not just CAD/code) is rare — study how data and strategy drove their design.
- Pairing CAD + code + strategy gives the most complete reproducibility picture in FRC.

CD thread: https://www.chiefdelphi.com/t/466792 · CD CAD, code & strategy release

★ Study pick	**1690 Orbit** Doppler ISR	**EPA pending**

Remarkably compact, world-class fast-cycle speaker shooter

SHOOTER (SPEAKER / AMP) Speaker shooter built into an unusually small envelope (judges/observers repeatedly noted how compact it is).	**INTAKE (NOTE)** NOTE intake tuned for a rapid, repeatable cycle (see CAD).
INDEXER / FEED Short feed path to keep the cycle fast (see CAD).	**DRIVETRAIN** Swerve.
VISION Unconfirmed (check CAD)	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- World-class results come from cycle-time obsession and tight packaging, not feature count.
- Study how much capability Orbit fit into a small, low robot.

Onshape CAD: https://cad.onshape.com/documents/065fb1a7f111b2de05ba208b/w/67f48a3bae8c68dc38e22239/e/e16d7c04df2851292695e2e5 · CD reveal — Doppler

★ Study pick	**118 Robonauts** Twister USA-TX	**EPA pending**

Highly complex shooter robot; CAD + code + tech binder + custom avionics

SHOOTER (SPEAKER / AMP) Speaker NOTE shooter; full subsystem breakdown documented in the 2024 Twister technical binder.	**INTAKE (NOTE)** NOTE intake (detailed in the tech binder).
INDEXER / FEED Feed/indexing path (tech binder).	**DRIVETRAIN** Swerve.
VISION Unconfirmed (check binder/code)	**PUBLIC CAD / DESIGN** CD CAD, code & binder release

LESSONS FOR STUDENTS

- Custom AVIONICS: Robonauts run their own IPDU power-distribution PCBs and a custom CAN node enabling a star CAN topology with optional on-board termination — a rare, advanced electrical study.
- Splitting a PCB into switching + breakout boards lets you iterate connector layout fast and cheap.
- 118 also authors the FRC Everybot — the canonical low-resource reference design.

CD thread: https://www.chiefdelphi.com/t/478131 · CD CAD, code & tech binder · GrabCAD — Twister

★ Study pick	8033 Highlander Robotics Banshee USA-CA	EPA pending

Carefully-architected speaker shooter; SFR finalist, champs Milstein finalist-captain

SHOOTER (SPEAKER / AMP) Two INDEPENDENTLY controlled sets of Stealth wheels create spin on the NOTE. A Maxspline cross-shaft carries virtually all torsional load so the shooter can be chain-driven from one side without twisting/flexing — and that same Maxspline doubles as the pivot (on a .75 extrusion with needle bearings) AND as a dead-axle roller the NOTE rides over.	**INTAKE (NOTE)** NOTE intake feeding the shooter (see CAD).
INDEXER / FEED Feed path into the shooter wheels (see CAD).	**DRIVETRAIN** Swerve.
VISION Open-source code (vision detail in repo).	**PUBLIC CAD / DESIGN** CD public CAD release

LESSONS FOR STUDENTS

- One Maxspline shaft serving as pivot + structural torsion member + note roller is an elegant part-count reduction worth studying.
- Two independently-controlled wheel sets let you impart deliberate spin for shot stability.
- Robustness came from a more carefully planned architecture / workflow than prior robots.

CD thread: https://www.chiefdelphi.com/t/469484 · CD public CAD release · Code — HighlanderRobotics/Crescendo

★ Study pick	2910 Jack in the Bot Typhoon USA-WA	EPA pending

Strong PNW shooter; CAD + code + tech binder, clone-friendly

SHOOTER (SPEAKER / AMP) Speaker NOTE shooter (full detail in the tech binder).	**INTAKE (NOTE)** NOTE intake (tech binder).
INDEXER / FEED Feed path (tech binder).	**DRIVETRAIN** Swerve.
VISION Unconfirmed (check binder/code)	**PUBLIC CAD / DESIGN** CD CAD, code & tech binder release

LESSONS FOR STUDENTS

- Documented failure mode: their climber gearbox put very high load on the motor — after Phoenix firmware added motor protection it stalled at low voltage; rolling back firmware caused ~180 A draw and a burned motor. A cautionary climber-design study.
- Publishing a clone-friendly CAD + Fabworks discount lowers the barrier for other teams to reproduce the robot.

CD thread: https://www.chiefdelphi.com/t/467192 · CD CAD, code & tech binder

★ Study pick	4738 Patribots Terry the T. Rex USA-CA	EPA pending

Requirements-driven shooter; standout technical binder

SHOOTER (SPEAKER / AMP) Speaker NOTE shooter (subsystem detail in the tech binder).	**INTAKE (NOTE)** NOTE intake (tech binder).
INDEXER / FEED Feed path (tech binder).	**DRIVETRAIN** Swerve.
VISION Unconfirmed (check binder/code)	**PUBLIC CAD / DESIGN** CD tech binder, CAD & code release

LESSONS FOR STUDENTS

- Their tech binder opens with an explicit REQUIREMENTS section — a great model for design-to-spec engineering.
- Released CAD + code + binder together; also explored dynamic autonomous (skipping to a different NOTE mid-path).

CD thread: https://www.chiefdelphi.com/t/467940 · CD tech binder, CAD & code · Onshape CAD

★ Study pick	3061 Huskie Robotics (2024) USA-IL	EPA pending

Open-Alliance, goals-first; full per-subsystem CAD + reusable libraries

SHOOTER (SPEAKER / AMP) Shooter released as its own CAD document (Shooter CAD).	**INTAKE (NOTE)** Intake released as its own CAD document (Intake CAD).
INDEXER / FEED Feed path documented alongside the subsystem CADs.	**DRIVETRAIN** Swerve — Drivetrain CAD plus 3061-lib, a swerve-focused starter library with advanced features.
VISION Unconfirmed (check code)	**PUBLIC CAD / DESIGN** CD 2024 Open-Alliance build thread

LESSONS FOR STUDENTS

- They publish each subsystem (climber/drivetrain/intake/shooter) as a separate CAD doc — easy to study one mechanism at a time.
- Reusable software: 3061-lib (swerve library) and SPOT (modular scouting app).
- 'Begin with the end in mind' — an explicit mission/objectives/goals process drives their season.

CD thread: https://www.chiefdelphi.com/t/447553 · CD 2024 build thread

★ Study pick	5940 BREAD Roti USA	EPA pending

Prototype-fast build; CAD + tech binder + ELECTRICAL binder + strong auto

SHOOTER (SPEAKER / AMP) Speaker NOTE shooter (detail in the tech binder).	**INTAKE (NOTE)** NOTE intake (tech binder).
INDEXER / FEED Feed path (tech binder).	**DRIVETRAIN** Swerve.
VISION Known for vision/odometry work (detail in code/binder).	**PUBLIC CAD / DESIGN** CD CAD & code release

LESSONS FOR STUDENTS

- They ship a dedicated ELECTRICAL binder in addition to the tech binder — an uncommon, high-detail wiring/electrical study.
- A largely wood-and-prototype-looking robot still produced a clean 6-piece auto — capability beats polish.
- 5940 is well known for vision/odometry libraries worth reading.

CD thread: https://www.chiefdelphi.com/t/466729 · CD CAD & code release

★ Study pick	6328 Mechanical Advantage (2024) USA-NH	EPA pending

Open-Alliance software powerhouse; authors of AdvantageKit

SHOOTER (SPEAKER / AMP) Speaker NOTE shooter (mechanism detail in the released CAD).	**INTAKE (NOTE)** NOTE intake (see CAD).
INDEXER / FEED Feed path (see CAD).	**DRIVETRAIN** Swerve.
VISION Advanced logging/vision via their own AdvantageKit framework (see code).	**PUBLIC CAD / DESIGN** CD 2024 Open-Alliance build thread

LESSONS FOR STUDENTS

- 6328 author AdvantageKit — the de-facto FRC logging/replay framework; their code is the reference for log-based development.
- A genuinely 'open book' build thread documents high-level structure decisions, not just the final robot.

CD thread: https://www.chiefdelphi.com/t/442736 · CD 2024 build thread

★ Study pick	2930 Sonic Squirrels Maestro USA-WA	EPA pending

Strong PNW shooter robot (reveal-documented)

SHOOTER (SPEAKER / AMP) Speaker NOTE shooter (mechanism detail in the released CAD).	**INTAKE (NOTE)** NOTE intake (see CAD).
INDEXER / FEED Feed path (see CAD).	**DRIVETRAIN** Swerve.
VISION Unconfirmed (check CAD)	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- A clean, competitive PNW-DCMP-level speaker robot to compare against the heavier-documented designs.
- Reverse-engineer the released Onshape to see how the cycle is packaged.

Onshape CAD: https://cad.onshape.com/documents/bbdb71fe91301c3d3bc1c45c/w/39c25dbe908c4708e0ef4712/e/29955289b046252b3d8a8198 · CD reveal — Maestro

★ Study pick	125 NUTRONs MOMENTUM USA-MA	EPA pending

Veteran NE team; bi-directional NOTE intake

SHOOTER (SPEAKER / AMP) Speaker NOTE shooter (mechanism detail in the released CAD).	**INTAKE (NOTE)** BI-DIRECTIONAL NOTE intake — can collect from either side, cutting the need to spin the robot before a pickup.
INDEXER / FEED Feed path to the shooter (see CAD).	**DRIVETRAIN** Swerve.
VISION Unconfirmed (check CAD)	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- A bi-directional intake removes a rotate-to-intake step — a nice cycle-time idea to study.
- Long-running veteran team with clean, conventional build practices.

Onshape CAD: https://cad.onshape.com/documents/c85e07d30d75286a0f0583ae/w/79297073f6715d470505a0bd/e/ff2e6f77f6015b5de0faea74 · CD reveal — MOMENTUM

Additional well-documented robots for study

Each has a strong public release (CAD, code, binder, or detailed reveal) and a clear design lesson. ★ marks the recommended study picks.

Sources: the Spectrum 3847 CAD Collection and team CAD releases (Onshape, GrabCAD), Chief Delphi reveal / CAD-release threads, and match data from The Blue Alliance. EPA shown is Local EPA (approx), independently computed — not official Statbotics. "Unconfirmed" means a detail was not publicly published, not that it is absent.